Project Plan

Project Information

Project Name: Truck Asset Matchmaking Platform (TAMP)

Project Type: Front-End MVP

Developer: Refilwe Sothoane

Technology Stack:

- React
- Vite
- Tailwind CSS
- React Router DOM
- Lucide React

Project Goal

The goal of this project is to develop a simplified Minimum Viable Product (MVP) of the Truck Asset Matchmaking Platform (TAMP). The application demonstrates how freight owners can connect with transporters through a digital marketplace while allowing administrators to monitor platform activity.

The project focuses on demonstrating the complete user journey using mock data without requiring a backend service.

Project Objectives

The application will allow users to:

- Register and log in.
- Select a user role.
- Post freight loads.
- Post available trucks.
- View match recommendations.

- Accept or reject matches.
- Generate digital confirmation.
- Track deliveries.
- Submit ratings.
- Allow administrators to monitor users, compliance and platform activity.

SDLC Approach

The project followed a simplified Software Development Life Cycle (SDLC) consisting of:

1. Planning
2. Analysis
3. Design
4. Development
5. Testing
6. Documentation
7. Presentation Preparation

Development Timeline

Day 1 – Planning & Analysis

Activities

- Read and discussed the Business Requirements Specification (BRS).
- Identified the three platform users:
  - Freight Owner
  - Transporter
  - Administrator
- Reviewed the mandatory functional requirements (FR-01 to FR-12).
- Planned the application structure and navigation.
- Created the Git repository.

Deliverables

- Requirements checklist

- Initial project structure

- Git repository

Day 2 – Design

Activities

- Designed low-fidelity wireframes.

- Planned application routing.

- Designed reusable dashboard components.

- Selected the technology stack.

- Planned the application folder structure.

Deliverables

- Wireframes

- Navigation structure

- Component design

Day 3 – Front-End Development

Activities

- Developed authentication pages.

- Implemented role selection.

- Built Freight Owner dashboard.

- Built Transporter dashboard.

- Built Admin dashboard.

- Created reusable layout and sidebar components.

Deliverables

- Working dashboards

- Responsive navigation

- Shared components

Day 4 – Core Features

Activities

- Developed Post Load page.
- Developed Post Truck page.
- Implemented Match Recommendations.
- Developed Digital Confirmation page.
- Developed Trip Tracking page.
- Implemented Ratings page.

Deliverables

- Complete freight workflow
- Complete transporter workflow

Day 5 – Administration Features

Activities

- Developed Users management page.
- Developed Compliance page.
- Developed Audit Trail.
- Developed Platform Analytics.
- Improved dashboard consistency.

Deliverables

- Complete Admin Console
- Platform monitoring pages

Day 6 – Testing & Improvements

Activities

- Tested application navigation.
- Verified all application routes.
- Tested forms using mock data.
- Improved responsiveness.
- Fixed UI inconsistencies.

- Corrected routing and import issues.

Deliverables

- Stable application

- Bug fixes

- Testing evidence

Documentation & Final Review

- Updated README.

- Completed Solution Overview.

- Completed Architecture documentation.

- Completed Requirements Traceability.

- Completed Testing Summary.

- Reviewed all assessment requirements.

- Prepared project for submission.

Deliverables

- Complete documentation

- Submission-ready project

Expected Outcome

At the end of the project, the application demonstrates the complete Truck Asset Matchmaking Platform workflow, including:

- User authentication

- Role-based dashboards

- Freight load management

- Truck management

- Match recommendations

- Digital confirmation

- Trip tracking

- Ratings

- Administrative management
- Audit trail
- Platform analytics

The solution satisfies the requirements of the simplified TAMP MVP using mock data and a responsive frontend interface.